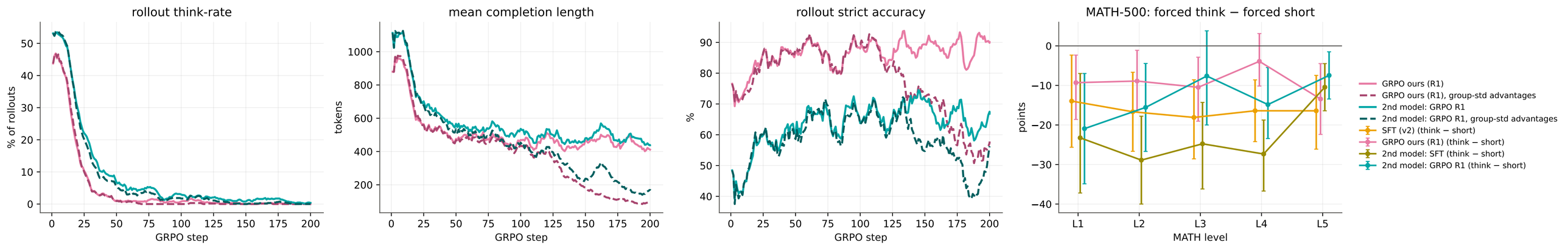
rollout think-rate
% of rollouts
GRPO step
mean completion length
tokens
GRPO step
rollout strict accuracy
%
GRPO step
MATH-500: forced think − forced short
points
MATH level
L1
L2
L3
L4
L5
GRPO ours (R1)
GRPO ours (R1), group-std advantages
2nd model: GRPO R1
2nd model: GRPO R1, group-std advantages
SFT (v2) (think − short)
GRPO ours (R1) (think − short)
2nd model: SFT (think − short)
2nd model: GRPO R1 (think − short)